

Implementation of ETL and ELT Big Data Pipelines for Medical Appointment Analysis

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Abstract

— This paper presents the design and implementation of a Big Data pipeline system for medical data analysis using two approaches: ETL (Extract, Transform, Load) and ELT (Extract, Load, Transform). The system processes data from two public datasets—No-Show Medical Appointments (110,527 rows) and Disease Symptoms & Patient Profile (~350 rows)—to identify patient no-show patterns and disease profiles. The ETL pipeline performs transformation using Python (Pandas) before loading, while the ELT pipeline loads raw data directly into a SQLite staging area and applies SQL-based transformations in-database. Both pipelines produce a Star Schema data warehouse with one fact table and three dimension tables. Key findings reveal a 20.2% no-show rate, with no-show patients waiting an average of 15.4 days compared to 8.7 days for those who attend. Weekend appointments show higher no-show rates (23.1%) than weekday appointments (20.2%). An interactive dashboard was developed using Looker Studio for visualization and analysis.

Keywords—Big Data; ETL Pipeline; ELT Pipeline; Star Schema; Medical Data; SQLite; Looker Studio; No-Show Prediction

I. INTRODUCTION

The healthcare sector generates massive volumes of data from various sources, including patient records, appointment scheduling systems, and clinical profiles. Efficient processing of this data is crucial for improving healthcare service quality, reducing operational inefficiencies, and enabling data-driven decision making.

Patient no-show at medical appointments is a significant problem in healthcare systems worldwide. No-shows lead to wasted medical resources, increased waiting times, and reduced healthcare capacity. According to the dataset used in this study, approximately 20.2% of scheduled appointments result in no-shows, representing a substantial operational burden.

This paper proposes and implements a comprehensive Big Data pipeline system that processes medical appointment data using two complementary approaches: ETL (Extract, Transform, Load) and ELT (Extract, Load, Transform). The system identifies patterns associated with patient no-show behavior and visualizes insights through an interactive dashboard.

II. DATASET

This study utilizes two publicly available datasets from Kaggle:

IV. FEATURE ENGINEERING

Eleven new features were engineered from the raw data to enrich the analytical capabilities of the data warehouse:

Feature	Source	Description
wait_days	No-Show	Days between scheduling and appointment
appt_day_of_week	No-Show	Day of week (0=Sunday)
appt_month	No-Show	Appointment month
is_weekend	No-Show	Weekend flag (1=Yes)
comorbidity_count	No-Show	Number of comorbid conditions
has_fever	Disease	Fever presence (1=Yes)
has_cough	Disease	Cough presence (1=Yes)
bp_encoded	Disease	Blood pressure: 0=Low, 1=Normal, 2=High
cholesterol_enc	Disease	Cholesterol level encoding
symptom_count	Disease	Total number of symptoms
age_norm	Both	Min-Max normalized age [0,1]

TABLE II. ENGINEERED FEATURES

V. DATA WAREHOUSE DESIGN

Dataset	Rows	Source
No-Show Appt.	110,527	Kaggle/Brazil
Disease Symptoms	~350	Kaggle

TABLE I. DATASET SUMMARY

The No-Show Appointments dataset contains 14 columns including PatientId, AppointmentID, Gender, ScheduledDay, AppointmentDay, Age, Neighbourhood, and medical condition flags (Hypertension, Diabetes, Alcoholism, Handicap, Scholarship). The Disease Symptoms dataset contains clinical profiles with Fever, Cough, Age, Blood Pressure, Cholesterol, and Outcome Variable columns.

III. SYSTEM ARCHITECTURE

The system architecture consists of five main components: data sources, ETL pipeline, ELT pipeline, data warehouse, and analytics dashboard. Both pipelines share the same data sources but differ fundamentally in their processing approach.

A. ETL Pipeline

The ETL pipeline follows the traditional approach of transforming data before loading it into the warehouse. The pipeline consists of three stages:

1) Extract: Raw CSV files are read from both sources and stored in a raw/ directory without any modification. Metadata including row count, column names, file size, and execution time is logged.

2) Transform: Data cleaning (duplicate removal, missing value imputation, IQR-based outlier detection), standardization (Min-Max normalization, binary encoding), and feature engineering (11 new features including wait_days, comorbidity_count, has_fever) are performed using Python/Pandas. Six data quality validation rules are applied.

3) Load: Cleaned data is loaded into SQLite using a Star Schema with one fact table (fact_medical) and three dimension tables (dim_source, dim_outcome, dim_age_group).

B. ELT Pipeline

The ELT pipeline differs fundamentally by loading raw data first, then transforming in-database using SQL:

Both pipelines produce a Star Schema data warehouse in SQLite. The Star Schema was chosen over Snowflake Schema due to simpler analytical queries (single JOIN), better OLAP performance, and easier comprehension for business users.

A. Fact Table

The fact_medical table contains 19 columns including foreign keys to all dimension tables, plus metric columns: age, age_norm, gender_code, outcome_flag, comorbidity_count, wait_days, appt_month, appt_day_of_week, is_weekend, symptom_count, has_fever, has_cough, bp_encoded, cholesterol_enc, and disease_name.

B. Dimension Tables

dim_source: Stores data source metadata (no_show, disease).

dim_outcome: Stores outcome labels (No-Show/Negative, Show/Positive).

dim_age_group: Stores five age groups: Children (0-12), Teenagers (13-17), Young Adults (18-35), Adults (36-59), Elderly (60+).

VI. RESULTS AND ANALYSIS

A. SQL Analytical Queries

Eight SQL analytical queries were developed and executed against both ETL and ELT warehouses, producing consistent results that validate the equivalence of both approaches:

Q#	Query	Key Result
Q1	Data per source	No-Show: 110,476; Disease: 350
Q2	Outcome distribution	Show: 79.8%, No-Show: 20.2%
Q3	Age per outcome	No-show avg: 37.8y, Show: 36.9y
Q4	Age group dist.	Adults (18-35) dominant
Q5	No-show per month	May: 20.8%, June: 18.5%
Q6	Comorbidity/age	Elderly: 0.87, Children: 0.11
Q7	Weekday vs weekend	Weekend: 23.1%, Weekday: 20.2%
Q8	Top diseases	Asthma(16), Osteoporosis(12)

TABLE III. ANALYTICAL QUERY RESULTS

B. Key Findings

Finding 1 – Wait Days as No-Show Predictor: No-show patients wait an average of 15.4 days between scheduling and appointment, nearly twice as long as patients who attend (8.7 days). This strong correlation suggests that scheduling optimization could reduce no-show rates.

1) Extract: CSV files are read with absolutely no preprocessing.

2) Load: Raw data is loaded directly into a Data Lake (CSV files) and SQLite staging tables (stg_no_show, stg_disease).

3) Transform (in-database): All transformations are performed using SQL statements including CREATE TABLE AS SELECT, CASE WHEN, JULIANDAY(), STRFTIME(), GROUP BY HAVING for deduplication, and ROW_NUMBER() OVER() for surrogate keys.

Finding 2 – Weekend Effect: Appointments scheduled on weekends show a 23.1% no-show rate compared to 20.2% on weekdays, indicating that weekend scheduling may require additional reminder interventions.

Finding 3 – Age-Comorbidity Correlation: Comorbidity count increases significantly with age, from 0.11 for children to 0.87 for elderly patients, suggesting age-stratified healthcare resource allocation.

Finding 4 – Disease Profile: Asthma (16 cases), Osteoporosis (12), and Stroke (11) are the most prevalent conditions in the Disease Symptoms dataset, informing healthcare resource prioritization.

VII. ETL VS ELT COMPARISON

Aspect	ETL	ELT
Order	Extract→Transform→Load	Extract→Load→Transform
Transform location	Python/Pandas (external)	SQL in-database
Staging area	None	stg_no_show, stg_disease
Scalability	Limited by memory	Database-optimized
Audit trail	Log files	Raw staging tables
Output table	fact_medical	fact_medical_elt

TABLE IV. ETL vs ELT COMPARISON

VIII. DASHBOARD

Two interactive dashboards were developed using Looker Studio, one for each pipeline (ETL and ELT). Each dashboard contains:

- KPI Cards: Total patients (110,821), No-show rate (20.2%), Average age (37.1y), Average wait days (10)
- Bar Chart: Monthly no-show rate trend (April-June 2016)
- Pie Chart: Outcome distribution (Show vs No-Show)
- Treemap: Top disease distribution
- Bar Chart: Average wait days per outcome

IX. CONCLUSION

This paper presents a complete Big Data pipeline implementation for medical appointment analysis using both ETL and ELT approaches. The system successfully processes 110,827 records from two heterogeneous data sources and produces a Star Schema data warehouse with 11 engineered features and 8 analytical SQL queries.

The key finding that no-show patients wait nearly twice as long as attending patients (15.4 vs 8.7 days) provides actionable insight for

healthcare administrators to implement targeted reminder systems for appointments with long lead times. Both ETL and ELT pipelines produce equivalent analytical results, validating the system's correctness and reproducibility.

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